

Agenda

↳ Matrix Factorization

↳ PCA

↳ SVD

→ MF → 3 classes

$$\rightarrow 6 = \overset{\checkmark}{\underbrace{2}} * \overset{\checkmark}{\underbrace{3}}$$

↳ factors

$$\rightarrow A_{n \times m} = \underbrace{B}_{n \times d} \underbrace{D}_{d \times m}$$

$$12 = 2 * 3 * 2$$

$$A_{n \times m} = B_{n \times d} C_{d \times d'} D_{d' \times m}$$

User-Item Matrix

$$A_{n \times m} = \begin{matrix} & I_1 & I_2 & I_j & \dots & I_m \\ \begin{matrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{matrix} & \begin{bmatrix} 5 & 2 & \downarrow & \dots & ? \\ ? & & \boxed{\text{shaded}} & \dots & ? \\ \vdots & & & & \vdots \end{bmatrix} \end{matrix}$$

Matrix completion $\rightarrow$ Matrix factorization

$$A_{n \times m} = B_{n \times d} C_{d \times m}^T$$

$$A_{ij} = B_i C_j^T$$

1×1 $1 \times d$ $d \times 1$

$A_{n \times m}$ =

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} I_1 & I_2 & I_j & I_m \\ 5 & 2 & \downarrow & ? \\ ? & & \boxed{\text{shaded}} & ? \\ \vdots & & & \vdots \end{bmatrix} \leftarrow \text{User-Item matrix}$$

$\xrightarrow{\quad d \quad}$

A_{ij}

$B =$
 $n \times d$

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \end{bmatrix}$$

$\rightarrow u_i$

User matrix in d-dim

$C_{m \times d} =$

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_j \\ I_m \end{bmatrix} \begin{bmatrix} | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \\ | & | & | & | & | \end{bmatrix}$$

$\rightarrow I_j$

Item matrix in d-dim

$$d < \min(m, n)$$

$$A_{|m} = B_i C_m$$

$$A = \begin{matrix} & I_1 & I_2 & \dots & I_m \\ \begin{matrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{matrix} & \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \end{matrix}$$

$$B_{n \times d} = \begin{matrix} & \xleftarrow{d} \xrightarrow{d} \\ \begin{matrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{matrix} & \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \end{matrix}$$

$$C =_{m \times d}$$

$$\begin{matrix} & \xleftarrow{d} \xrightarrow{d} \\ \begin{matrix} I_1 \\ I_2 \\ \vdots \\ I_m \end{matrix} & \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \end{matrix}$$

$$\begin{array}{c} u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} I_1 \quad I_2 \quad I_3 \quad I_4 \\ \left[\begin{array}{cccc} 5 & ? & 3 & 1 \\ 4 & ? & ? & 1 \\ 1 & 1 & ? & 5 \end{array} \right] \end{array} \xrightarrow{\text{red arrow}} X$$

$A_{3 \times 4}$

$$\begin{array}{c} \rightarrow -u_1 \\ u_2 \\ u_3 \end{array} \begin{array}{c} \downarrow \quad \downarrow \\ \left[\begin{array}{cc} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{array} \right] \end{array}$$

$B_{3 \times 2}$
↑
d

$$\begin{array}{c} \rightarrow \\ \rightarrow \end{array} \begin{array}{c} I_1 \quad \downarrow I_2 \quad I_3 \quad I_4 \\ \left[\begin{array}{cccc} c_{11} & c_{21} & c_{31} & c_{41} \\ c_{12} & c_{22} & c_{32} & c_{42} \end{array} \right] \end{array}$$

$C^T_{2 \times 4}$
↑
d

$$\left[\begin{array}{l} \checkmark \quad 5 = b_{11} c_{11} + b_{12} c_{12} \\ \checkmark \quad 3 = b_{11} c_{31} + b_{12} c_{32} \\ \quad \quad 1 = b_{11} c_{41} + b_{12} c_{42} \end{array} \right.$$

→ To calculate at test time

$$\checkmark \quad X = b_{11} \cdot c_{21} + b_{12} c_{22}$$

$$A = \begin{matrix} & I_1 & I_2 & I_3 & I_4 \\ u_1 & \boxed{5} & \boxed{?} & \boxed{3} & \boxed{1} \\ u_2 & \underline{4} & \boxed{?} & ? & 1 \\ u_3 & 1 & 1 & ? & 5 \end{matrix}$$

→ X

$$B = \begin{matrix} & u_1 & u_2 \\ u_1 & 1.77 & 1.00 \\ u_2 & 1.37 & 0.88 \\ u_3 & -0.60 & 2.05 \end{matrix}$$

$$\hat{A} = BC^T = \begin{bmatrix} 4.98 & 2.34 & \boxed{2.99} & 0.99 \\ \underline{\underline{3.98}} & 1.892 & 2.34 & 0.95 \\ 1.1 & 1.07 & 1.35 & 4.96 \end{bmatrix}$$

$$C = \begin{matrix} & I_1 & I_2 \\ I_1 & 2.14 & 1.20 \\ I_2 & 0.88 & 0.78 \\ I_3 & 1.13 & 0.99 \\ I_4 & -0.69 & 2.22 \end{matrix}$$

→ $d=25$

① → Randomly initialize B & C matrix
 $n \times d$ $m \times d$

② $\begin{cases} \min_{B_i, C_j} \sum (A_{ij} - B_i C_j^T)^2 \\ \text{s.t. } A_{ij} \neq ? \end{cases}$

B_i 

C_j^T 

ALS $\min (A_{ij} - B_i C_j^T)^2$

↳ fix B_i & update C_j ✓

↳ fix C_j & update B_i ✓

→ Faster

→ Cold Start

A

Item - Item

$$A = \begin{matrix} & I_1 & I_2 & I_3 & I_4 \\ u_1 & \begin{pmatrix} 5 \end{pmatrix} & \begin{pmatrix} ? \end{pmatrix} & \begin{pmatrix} 3 \end{pmatrix} & \begin{pmatrix} ? \end{pmatrix} \\ u_2 & \begin{pmatrix} 4 \end{pmatrix} & \begin{pmatrix} ? \end{pmatrix} & ? & ? \\ u_3 & \begin{pmatrix} 1 \end{pmatrix} & ? & ? & ? \end{matrix}$$

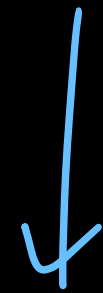
→ X

$$= B C^T$$

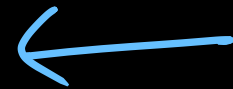
→ Break until 22:21 PM

PCA

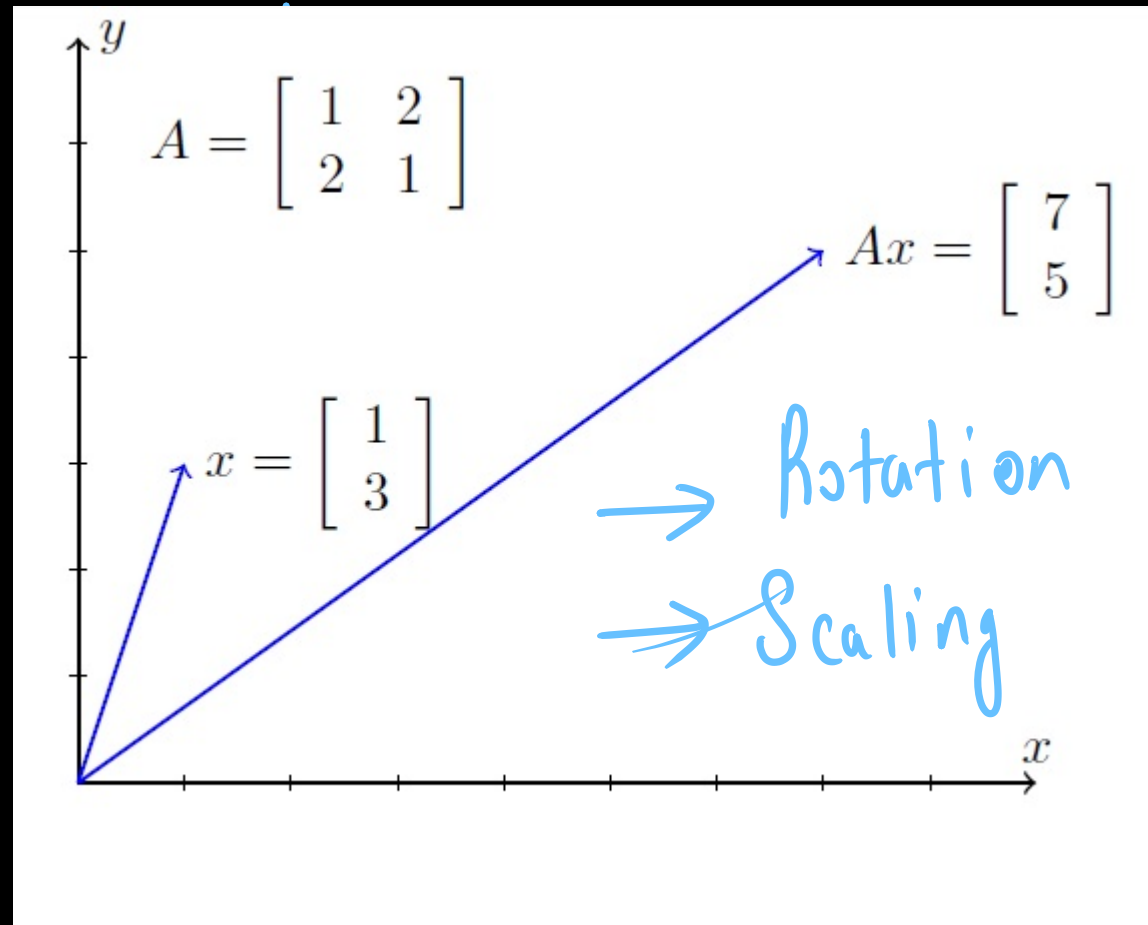
$X_{n \times d} \rightarrow X_{std} \rightarrow \text{Find covariance } X^T X$



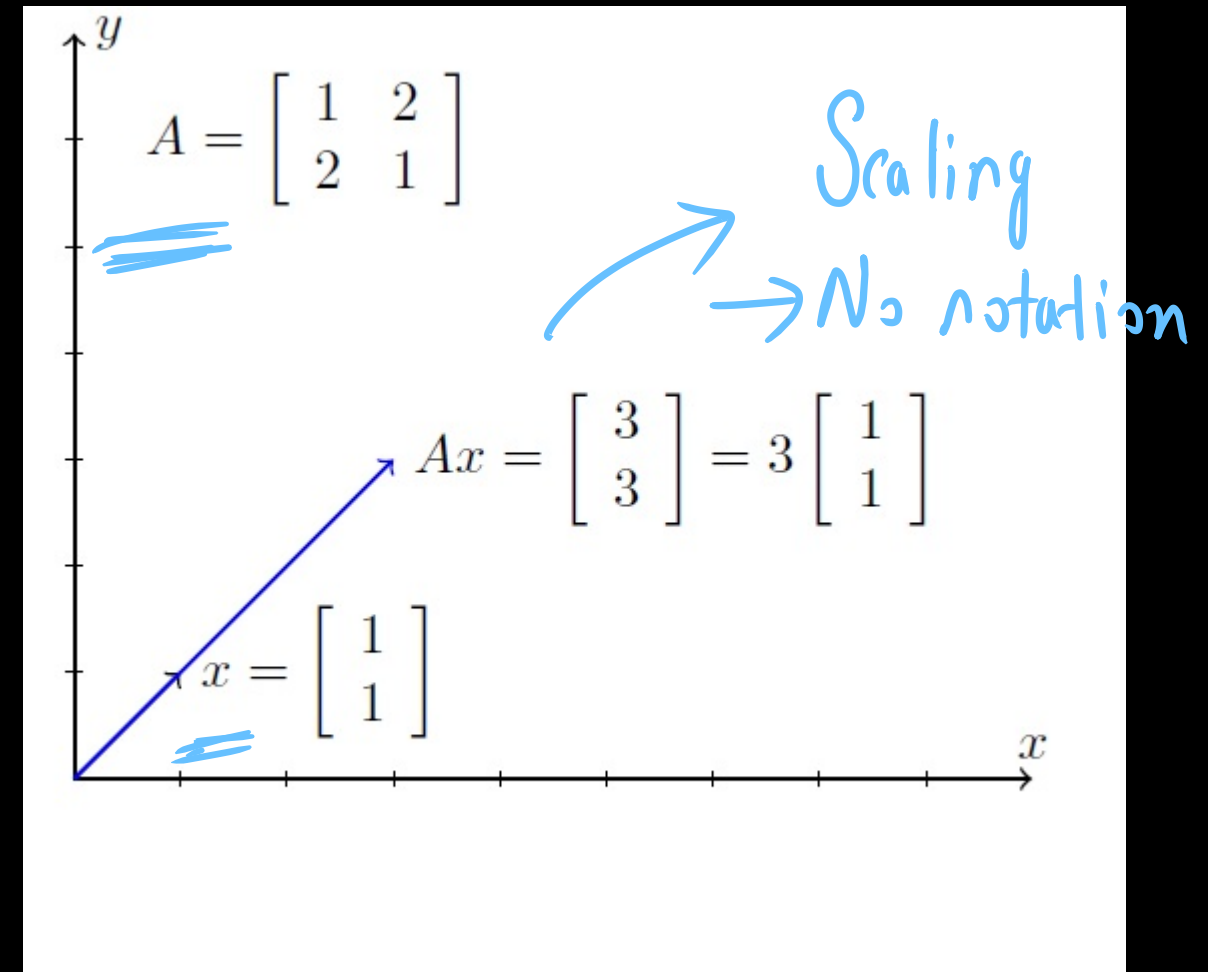
Get Eigen value & Eigen Vector
of $X^T X$



Project X
on top- K
eigen vector
corresponding to
top- K eigen values



Ax ✓



$$Ax = \lambda x$$

$$A v_1 = \lambda_1 v_1$$

$$A v_2 = \lambda_2 v_2$$

⋮

$$A v_K = \lambda_K v_K$$

$$A \begin{bmatrix} \uparrow v_1 & \uparrow v_2 & \cdots & \uparrow v_K \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} \lambda_1 \uparrow v_1 & \uparrow & & \uparrow \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 6 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$V = \begin{bmatrix} \uparrow v_1 & \uparrow v_2 & \cdots & \uparrow v_K \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

$$= \begin{bmatrix} \uparrow v_1 & \uparrow v_2 & \cdots & \uparrow v_K \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_K \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ & & \ddots \\ 0 & & & \lambda_K \end{bmatrix}$$

$A V = V \Lambda$

$$AV = V\Lambda$$

$$\boxed{A_{d \times d} = V \Lambda V^{-1}}$$

→ Eigen Value Decomposition

↳ Exist only for
Square matrix

$$= V \Lambda V^T$$

↓ ↳ Eigen value of A

Eigen vector of A

$$X_{m \times d} \rightarrow X_{d \times d}^T X$$

$$\boxed{X_{d \times d}^T X = V \Sigma V^T} \rightarrow$$

$$\rightarrow A = X^T X_{d \times d} = V \Lambda V^T$$

V = Eigen vector of $A (X^T X)$

Λ = Diagonal Matrix containing eigen value of $A (X^T X)$

$\rightarrow$

$$A = B C^T$$

$$A = V \Sigma V^T$$

$$B = V \Sigma$$

$$C = V$$

→ SVD (Singular Value decomposition)

→ EVD nequive square matrix

$$\left(\underset{m \times d}{X} \rightarrow \underset{\underline{\underline{d \times d}}}{X^T X} \right)$$

$$A = V \Sigma V^{-1} \checkmark \checkmark \\ = V \Sigma V^T \checkmark \checkmark$$

$$A = U \Sigma V^T$$

$$\begin{array}{c}
 \xrightarrow{\quad} A_{n \times d} \quad \xrightarrow{\quad} \underbrace{\quad}_{\text{Rectangular}} \\
 \downarrow \\
 \text{Rectangular}
 \end{array}
 \quad
 \begin{array}{c}
 \xrightarrow{\quad} \underbrace{\quad}_{\text{Figen vector of } A^T A} \\
 \downarrow \\
 \text{Figen vector of } A^T A
 \end{array}
 \quad
 \begin{array}{c}
 \xrightarrow{\quad} \underbrace{\quad}_{\text{Figen vector of } AA^T} \\
 \downarrow \\
 \text{Figen vector of } AA^T
 \end{array}$$

$$\xrightarrow{\quad} A = BC^T \rightarrow M_f$$

$$\boxed{B = U \Sigma \quad C = V}$$

